Generalising Deformable Object Manipulation by Learning to Learn from Demonstrations

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***Abstract*—Research in deformable object manipulation interleaves probabilistic modelling, planning and control to realise robots that can solve complex tasks with highly uncertain object dynamics. These tasks range from shaping ribbons and typing knots to wrapping objects and preparing food. Past research has focused on modelling the geometry and dynamics of deformable objects for planning and control, but some recent techniques have circumvented the challenging object modelling problem by using learning from demonstration (LfD) techniques, like behavioural cloning. While promising, many of these methods fail to scale to real world environments because they demand new training data to generate policies for new task settings. This work emphasises the need to develop methods that quickly learn and transfer the structure of a task for application to human-in-the-loop systems in the real world.**

***Index Terms*—deformable object manipulation, meta-learning, optimal control**

I. Introduction

Deformable objects pose a significant and unavoidable challenge for robots that interact with real world environments. Even in simple settings, robots interfacing with ropes and cables are severely limited in the complexity of tasks that they can perform. Some of today's robots are able to do table-top knot tying, rope shaping, and cable routing, but these quasi-static tasks fall short of capturing the richness of simple deformable object manipulation (DOM) that humans are capable of doing. For example, a sailor can tie an elaborate set of functional knots to secure sails and anchor their boat to a dock. A rock climber can tie a figure-eight follow-through in their hand to secure their weight on the sharp end of a rope. Most people can manage to untangle the bundle of wired earbuds they've retrieved from their pocket or bag. Somehow humans easily negotiate these complex systems and can apply the appropriate actions to achieve functional object configurations. While it is a dream to build autonomous robots that are capable of replicating these tasks, the development of autonomous DOM for assistive robotics is essential.

Assistive robotics encapsulates many human-in-the-loop robotic scenarios where robots do tasks that are difficult for, or harmful to, the human. For example, a robotic system tasked with tying a person's shoe may require an assortment of looping motions (for knotting, cinching and velcro-ing) that share a common structure across different lace and shoe types. Similarly, fastening wearable devices (e.g., watches, arm bands and the like) may share common structural components such as winding, tensioning and flexing the object against a relatively rigid object. Many of robots today are capable of learning policies for parts of these tasks, but the availability of human interaction makes Learning from Demonstration (LfD) an appealing catalyst for learning complex tasks. A human caregiver, for instance, could train a robot to tie shoes in a variety of ways [1, 2]. However, learning skills from the caregiver's examples that can generalise to the deployed setting is a non-trivial problem; e.g., the training examples are not representative of the multitude of shoe lace shapes, lengths, materials and shoe types at the patient's home. Techniques like behavioural cloning often have inadequate generalisation performance in these scenarios because they rely on strong visual and physical correspondences between the training and deployed environment [17, 10, 14, 19]. These scenarios demand, instead, systems that can learn robust policies from small amounts of data and apply them to new environments. Meta-learning may offer a way to develop such systems.

Motivated by a need to generalise demonstration-based task execution, this work investigates how meta-learning can provide a way to quickly learn latent task descriptions from demonstration data. These task representations can contribute to the objectives in motion planning and optimal control.

II. Related Work

1) Planning for rope and cloth manipulation: A large body of work in manipulation planning has developed a vast collection of geometric representations for ropes, cables and cloth. Graphical and topological state descriptions were successfully used for planning complex knotting [11, 6]. Lower-level representations took into account the dynamical properties of ropes, cloth and 3D deformable objects using linear [23, 29, 22] and mesh mass-spring models [18]. Deep learning has contributed fantastically rich geometric models [19] that enabled grasp point selection [5]. Many of these representations include deformation models that pick-and-place planners used to solve quasi-static tasks. Some methods addressed more dynamic

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tasks like high-speed knot tying, dynamic cloth unfolding and "in air" knotting [26, 7, 16].

2) Learning from demonstration (LfD) for rope and cloth manipulation: LfD techniques have become common tools for improving the efficacy of many rope manipulation algorithms because they obviate the need to develop rich object models. However, much of this research does not prioritise the need for learning policies that transcend dramatic dynamical changes [14, 10, 17]. This need may motivate the desire to learn policies that leverage simpler task models that have good generalisation performance on tasks with mutable object properties.

3) Rope and cloth manipulation sans LfD: Recent methods have achieved impressive results on rope shaping and cloth folding tasks without guidance from human demonstration [25, 28, 23, 12, 3] , yet it is unclear how effectively they generalise to similar tasks with dramatically different dynamics. Since these data-driven approaches rely on highly over-parameterised models that, when unrestricted, may encode task-irrelevant features, the learnt models may prevent them from performing well on, say, a paper folding task. Moreover, these approaches do not seem to yield representations that encode the task and dynamics in a selectively transferable manner.

4) Leveraging task abstractions for DOM - a knotting example: Several knotting theoretic approaches succeeded in tying and untying of ropes with and without LfD. Wakamatsu et al. posed the knotting problem as a graph search optimisation over topological states. Their planner could achieve arbitrarily complex knots provided a rich library of topological descriptions of actions and sub-actions [22]. Recently, Yan et al. addressed the knot tying problem by implementing an RL-learning algorithm that generated motion sequences for primitives prescribed by knotting theory; the RL algorithm was seeded by demonstrations of the abstract actions [27]. Grannen et al. took a look at a slightly different task; untying. They developed a supervised learning algorithm which identified task relevant visual features for unknotting. They used a geometric planner to select abstract action sequences and the perception module to infer key points on which to localise the pick, hold and pull actions [5]. Evidently, complex long-horizon rope knotting and unknotting is solvable by hierarchical motion planning algorithms. Knotting theory provided key dynamics-invariant structural elements for the task.

In thinking about tasks beyond knotting ropes, there do not appear to be intuitive analogues that parallel topological states and actions that can inform policy optimisation for DOM. LfD approaches, on the other hand, may expedite the process of conveying task structure across controllers if meta-learning is useful for learning how to extract such an abstraction from demonstrations.

5) Meta-learning: The meta-learning problem formulation in Nagabandi et al. provides a lens through which to view this task modelling problem [13]. It considers the problem of online adaptation of dynamics models in the context of model-based RL. However, our meta-learning algorithm aims to learn a motion model from demonstrations of how to do object-specific tasks.

III. Problem Formulation

The primary goal is to derive, from demonstration data $\tau_N = \{\mathbf{z}_0, \ldots, \mathbf{z}_{N-1}\}$, an abstract representation of a core task that is independent of any particular object's dynamics or geometry. Let $M_\theta(\mathbf{a}) = \arg\max_{\mathbf{a}'} p_\theta(\mathrm{X}_{\mathrm{n}+1} = \mathbf{a}' | \mathrm{X}_{\mathrm{n}} = \mathbf{a})$ be this abstract representation, where random variables $\mathrm{X}_{\mathrm{n}+1}, \mathrm{X}_{\mathrm{n}} \in \mathbb{R}^m$ are sequential actions and θ are the motion model's parameters. Figure 1 shows examples of such core tasks. A necessary assumption is that it is feasible to model the task-specific motion patterns independently of the object characteristics (see Sec. VI). If that is the case, then during task execution, a policy can leverage this abstraction to perform the core task with new deformable objects; provided object models learnt offline. This proposed approach will apply meta-learning to learn M_θ for optimal control of systems with complex non-linear dynamics.

A. Recurrence-based meta-learning

Given a distribution of tasks $P(\mathcal{T})$ with shared structure, we seek a meta-learning algorithm that learns meta-parameters ϕ and model parameters θ for a learning algorithm $A_{\theta,\phi} = A_\phi(\mathcal{D}_\mathcal{T}, \theta)$ such that it performs well on a dataset $\mathcal{D}_\mathcal{T} \sim P(\mathcal{T})$ with high probability. Typically, a meta-learner seeks to minimise the expected (supervised) loss over P. That is,

$$\min_{\theta,\phi} \mathbb{E}_{\mathcal{D}_\mathcal{T}^{\text{train}} \sim P} \left[\mathcal{L}_\theta(\mathcal{D}_\mathcal{T}^+, A_\phi(\mathcal{D}_\mathcal{T}^-, \theta)) \right] \tag{1}$$

where $\mathcal{D}_\mathcal{T}^- \sim \mathcal{D}_\mathcal{T}^{\text{train}}$ denotes the training set and $\mathcal{D}_\mathcal{T}^+ \sim \mathcal{D}_\mathcal{T}^{\text{train}}$ denotes test set. Both sets are drawn without replacement from the meta-training dataset. After training, the learner A_{ϕ^*,θ^*} should quickly learn the tasks in the meta-test dataset $\mathcal{D}_\mathcal{T}^{\text{test}}$. Following Duan et al. (2016), we will use a recurrent model to encode $A_{\phi,\theta}$, yielding a recurrence-based meta-learning problem [4]. In this case, the parameters ϕ, θ represent the weights for the hidden state and the corresponding update weights, respectively.

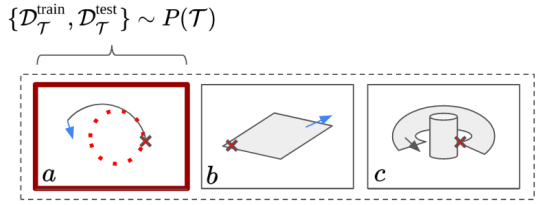


Fig. 1. A cartoon illustration of candidate task distributions for testing this approach. We propose 2D shaping, tensioning and winding as core tasks, but experiments will focus on the shaping task. Each distribution will consist of object specific sub-tasks; i.e., the core task will be done with an assortment of objects that vary shape, stiffness, etc. a) 2D shaping: a deformable object must be configured into a target shape, indicated by the red dots. One region of the object will be fixed to the plane (see red 'x'). b) 3D tensioning c) maintaining contact while winding about a rigid cylinder.

IV. Approach

A deformable object task distribution consists of many tasks that share an underlying core task (see Figure 1). The core task has a fixed goal and it may involve interaction between a rigid and a deformable object. The differences between the tasks in the distribution result from morphological changes to the deformable object. For example, a rope from one task may undergo dramatic shift in stiffness, change in elasticity, dimension scaling, etc. to yield another task in the distribution. In the latter case, the rope may no longer resemble a member of the ropes object class, but the bands object class instead. We call this hypothetical task distribution $P(\mathcal{T})$. Demonstration data will represent policies that solve tasks sampled from $P(\mathcal{T})$. Our experiments will center on the 2D shaping task, but this approach could be evaluated on the 3D tensioning and a winding task in the future.

A. Environment modelling

Efficient optimal control algorithms leverage dynamics models to make predictions about the future when selecting control inputs. These models typically work well with low-dimensional state representations. In the case of deformable object manipulation, self-supervised algorithms can learn low-dimensional image encodings and corresponding dynamics models by training neural models, denoted by M_ω and M_ν, respectively [28]. Specifically, this work parameterises the dynamics with a MLP $\mathrm{M}_\nu \rightarrow f_\nu(\mathbf{x}, \mathbf{u}) \approx \mathbf{x}'$ and the image encoder with a seven-layer convnet $\mathrm{M}_\omega \rightarrow g_\omega(o) \approx \mathbf{x}$, where o is an image observation.

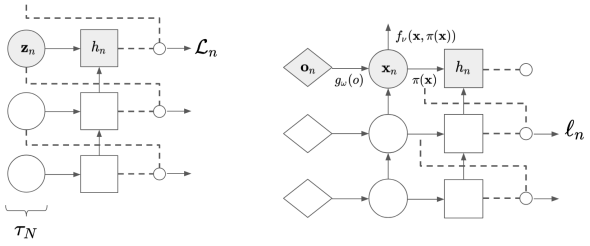


Fig. 2. Training the motion model p_θ from a single demonstration sequence $\tau_N = \{\mathbf{z}_0, \ldots, \mathbf{z}_{N-1}\}$ according to loss Eq. 2 (left). The hidden layers of the recurrent model are denoted by h_n. Optimising a policy $\pi(\mathbf{x})$ using instantaneous cost $\ell_n(\mathbf{x}, \mathbf{u}; \theta)$ (right).

B. Task modelling and execution

Sec. III outlined a meta-learning problem for learning a representation of a core task. Figure 2 summarises the pipelines for task modelling and optimal control, respectively. From a set of demonstration sequences $\tau_N \sim \mathcal{D}_\mathcal{T}$, $A_{\phi,\theta}$ will learn the dynamics-invariant motion model M_θ according to the maximum likelihood objective. Specifically, this objective will take the form of Eq. 1 with loss,

$$\mathcal{L}_\theta(\mathcal{D}_\mathcal{T}, \theta') = -\frac{1}{N}\sum_{n=0}^{N-1} \log p_{\theta'}(\mathrm{X}_{n+1} = z_{n+1} | \mathrm{X}_n = z_n) \quad (2)$$

where $\theta' = A_\phi(\mathcal{D}_\mathcal{T}^-, \theta)$ and $\mathcal{D}_\mathcal{T} = \mathcal{D}_\mathcal{T}^+$. The motion model will be incorporated into the cost function of a locally optimal control algorithm [24]. Formally, provided a goal configuration $\mathbf{x}_g$, the controller $\pi(\mathbf{x}) = p(\mathbf{u}_n|\mathbf{x}_n)$ will optimise the objective $\min_\pi J_\pi(\mathbf{x}_0)$ with instantaneous cost $\ell_n(\mathbf{x}, \mathbf{u}; \theta) = \lambda_1 \|\mathbf{x}_g - \mathbf{x}_n\|_2^2 + \lambda_2 \|\mathbf{u}_n - M_\theta(\mathbf{u}_{n-1})\|_2^2$,

$$J_\pi(\mathbf{x}_0) = \mathbb{E}_{\hat{\tau}\sim p(\hat{\tau}|\pi)}\left[\lambda_1 \|\mathbf{x}_g - \mathbf{x}_0\|_2^2 + \sum_{n=1}^{N-1} \gamma^n \ell_n(\mathbf{x}, \mathbf{u}; \theta)\right]$$

where $\mathbf{x}$ is given by $\mathrm{M}_\omega(\mathbf{o}_n)$, $\lambda_{1,2}$ are scalar constants and

$$p(\hat{\tau} \mid \pi) = p(\mathbf{x}_0) \prod_{n=0}^{N-1} p(\mathbf{x}_{n+1}|\mathbf{x}_n, \mathbf{u}_n)\pi(\mathbf{x}).$$

V. Experiment Setup

Preliminary environments for training object models for simulated rope and cloth are available as custom Deep Mind Control environments. The simulations run using MuJoCo physics engine [21, 20]. In one example, a one-step MPC controller applies these models to attempt to unravel a rope from a cylinder in simulation (see Figure 3). The cost function currently does not depend on the demonstration data, but the plan is to integrate the learnt task model as a component of the cost for the shaping task.

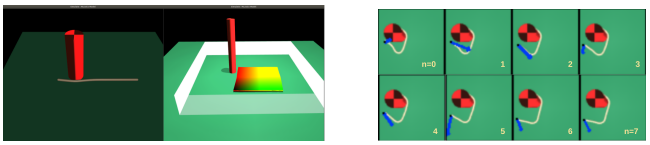


Fig. 3. Two simulated settings for the (rigid-object) wrapping core task with rope and cloth (Left). Planning with learnt dynamics using one-step MPC for a rope unwinding task (Right). The blue vectors indicate the pulling location and direction for a selected action.

VI. Open Questions

One underlying assumption in this work is that it is possible to decouple the task description from environment dynamics. By the success of [27], it seems possible - at least in the context of rope tying tasks. There may be open questions about whether 1) this assumption holds for DOM and 2) how learners can extract these invariances. The work done by [8] seeds hope in the idea that end-to-end models are partially adaptable for new environments. Under a similar assumption they show that it is possible to adapt learnt controllers in modular ways, e.g., using self-supervised learning objectives to adapt the network parameters associated with dynamics prediction.

VII. Conclusion

This paper highlights the problem of generalising policies that are learnt by demonstration in the deformable object manipulation task domain. The proposed work aims to design and implement a meta-learning algorithm that will enable fast task learning in environments where the characteristics of the deformable object are likely to change.

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